AI for Redistricting — Spring 2026

Gerrymandering Analysis of Massachusetts Congressional Districts

2024 Presidential & Senate Elections | ReCom Ensemble Analysis

University of San Francisco • CS 463

What is Redistricting?

Every 10 years, after the U.S. Census, congressional district boundaries are redrawn to reflect population changes.

Who draws the maps?

In most states, the state legislature draws congressional maps — creating a conflict of interest when the party in power draws lines to favor itself.

What must maps satisfy?

Equal population, contiguity, compactness, and compliance with the Voting Rights Act (VRA).

Why is it complicated?

There are exponentially many valid maps. Determining whether a map is 'fair' requires statistical comparison to an ensemble of alternatives.

Gerrymandering Techniques

Packing

Concentrating opposition voters into a few districts so they win big but waste votes.

Cracking

Splitting opposition voters across many districts so they can't form a majority anywhere.

Hijacking

Redrawing lines to put two incumbents in the same district.

Massachusetts Context

9
Congressional Districts

7.03M
Total Population (2020)

100%
Dem-held seats (current)

2023
Current map enacted

Map Creation

The current Massachusetts congressional map was enacted in 2023 by the Democratic-controlled state legislature. Massachusetts does not use an independent redistricting commission.

Partisan Landscape

Massachusetts is a heavily Democratic state. Democrats hold all 9 congressional seats. In 2024, Harris won approximately 64% of the statewide presidential vote, and the Democratic Senate candidate won by a similar margin.

Legal Landscape

Massachusetts has not faced significant legal challenges to its congressional map. The state's congressional delegation is entirely Democratic, raising questions about whether the map constitutes a partisan gerrymander favoring Democrats.

Racial Landscape

Massachusetts is majority white (~71%), with significant Black, Hispanic, and Asian populations concentrated in Boston and surrounding areas. District 7 (Boston-centered) has historically served as a majority-minority district.

Data & Methods

Data Sources

1 **2020 Census Blocks**

107,278 census blocks with total population (P0010001). Used to aggregate population to precinct level.

2 **MA County Shapefile**

14 counties. Used to nest precincts within counties during MAUP repair.

3 **2024 General Election Precincts**

2,383 precincts with vote totals for Presidential (Harris/Trump) and Senate races.

MAUP Data Cleaning Pipeline

1 **Reproject**

Aligned all three shapefiles to UTM Zone 19N for accurate spatial operations.

2 **smart_repair**

Fixed 519 overlaps, 77 holes, and invalid geometries in election precincts, nesting within county boundaries.

3 **Rook → Queen**

Converted adjacencies < 30m to point adjacencies so GerryChain treats them as proper neighbors.

4 **Assign Population**

Used maup.assign() to map 107k census blocks → 2,383 precincts and sum population.

5 **Validation**

maup.doctor() returns True. Block total = Precinct total = 7,029,917.

Ensemble Method: ReCom

How ReCom Works

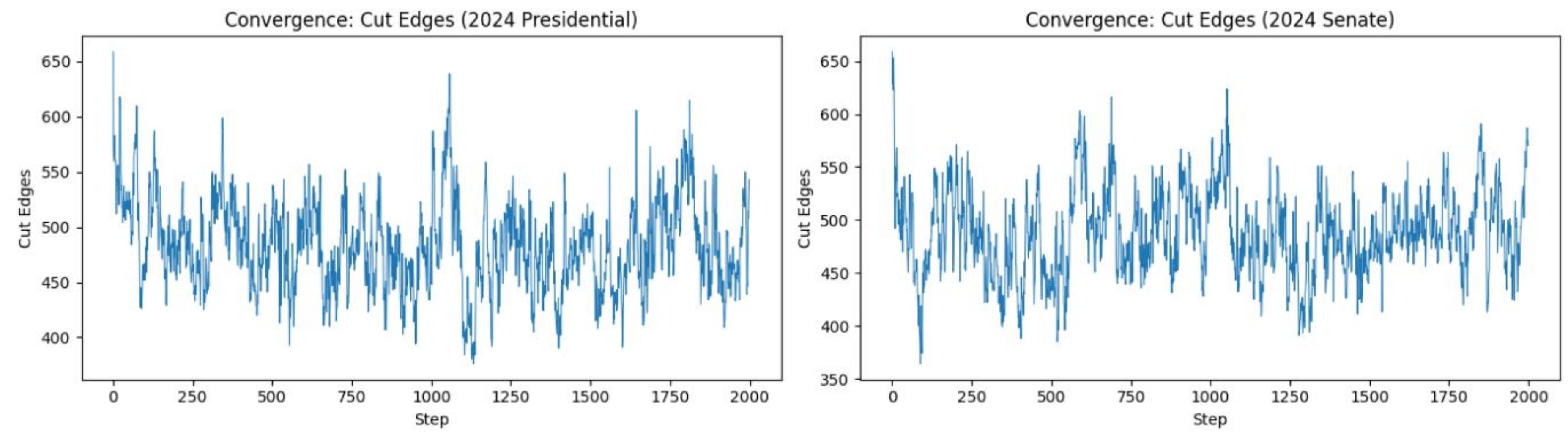
1. Pick two adjacent districts at random.
2. Merge them into one region.
3. Build a random spanning tree of the merged region.
4. Cut one edge of the spanning tree to create two new valid districts satisfying population constraints.
5. Repeat 2,000 times to generate the ensemble.

Parameter Choices

Steps per chain	**2,000**
Number of chains	**2 (Pres + Senate)**
Population tolerance	**±2%**
Node repeats	**2**
Compactness bound	**≤ 2× initial cut edges**
Contiguity	**Enforced**
Accept function	**Always accept**
Graph adjacency	**Queen**

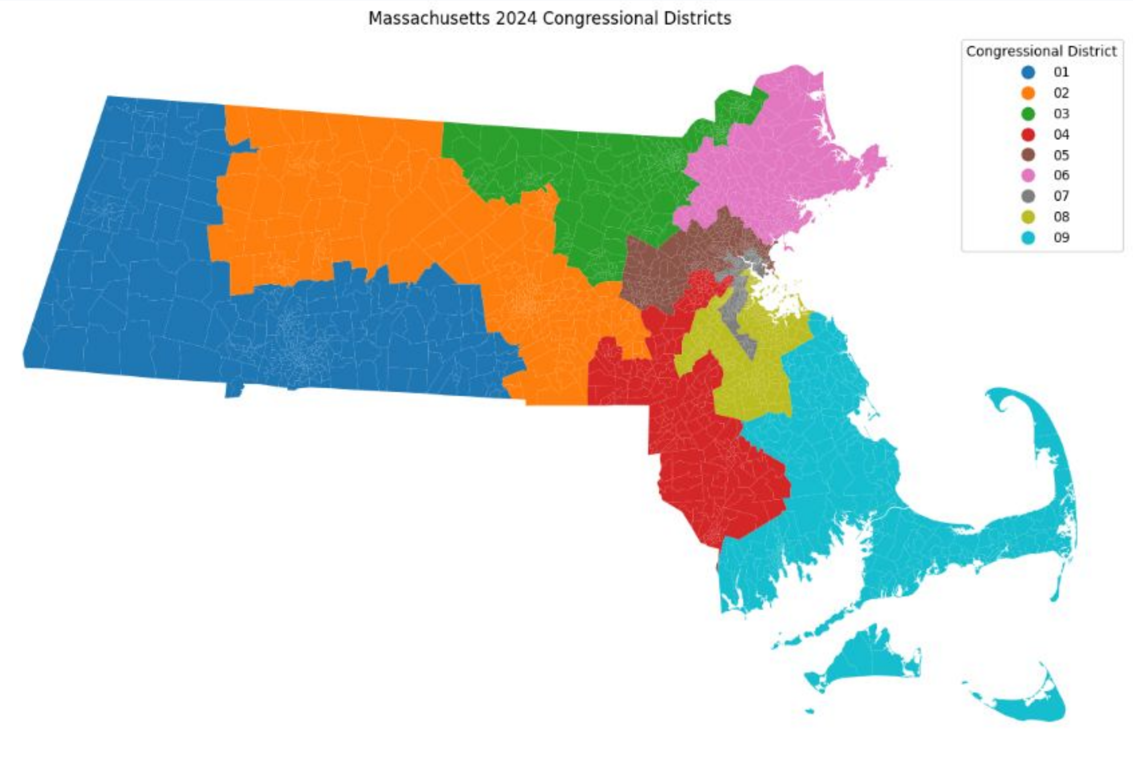
Convergence Evidence

The cut edges trace for both chains fluctuates randomly around a stable mean with no visible trend or drift — confirming the chain mixed well and 2,000 steps was sufficient.



✓ No drift observed • Chain stabilizes within first 200 steps • Both elections show consistent behavior

The Actual Massachusetts Map



9 districts
All Democrat-held

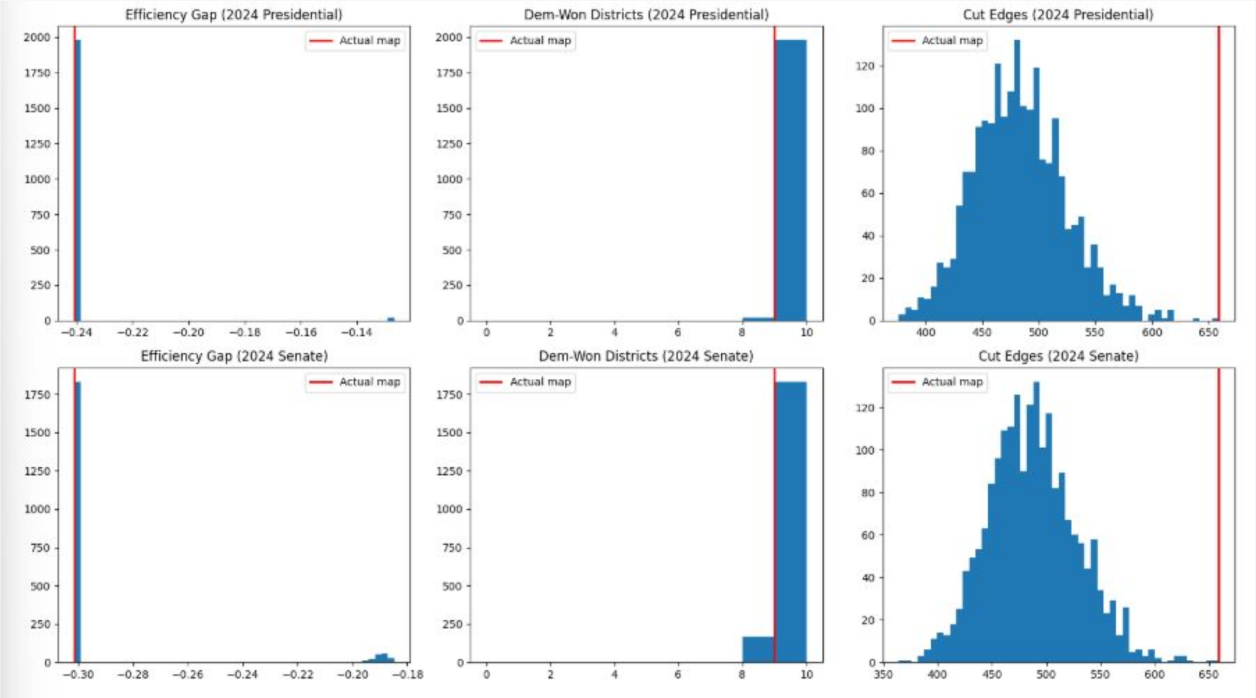
659
Cut edges in actual map

-0.24
Efficiency Gap (Pres)

-0.30
Efficiency Gap (Senate)

Outlier Analysis: Histograms

Red line = actual MA map value. Each histogram shows where the MA map falls relative to 2,000 randomly generated alternative maps.



Key finding: The actual map's Efficiency Gap sits in the extreme left tail of both distributions, and all 9 Democrat wins is a rare outcome in the ensemble.

Understanding the Efficiency Gap

What is the Efficiency Gap?

The Efficiency Gap measures wasted votes — votes cast for the losing party (all wasted) plus votes the winning party cast beyond what was needed to win (also wasted).

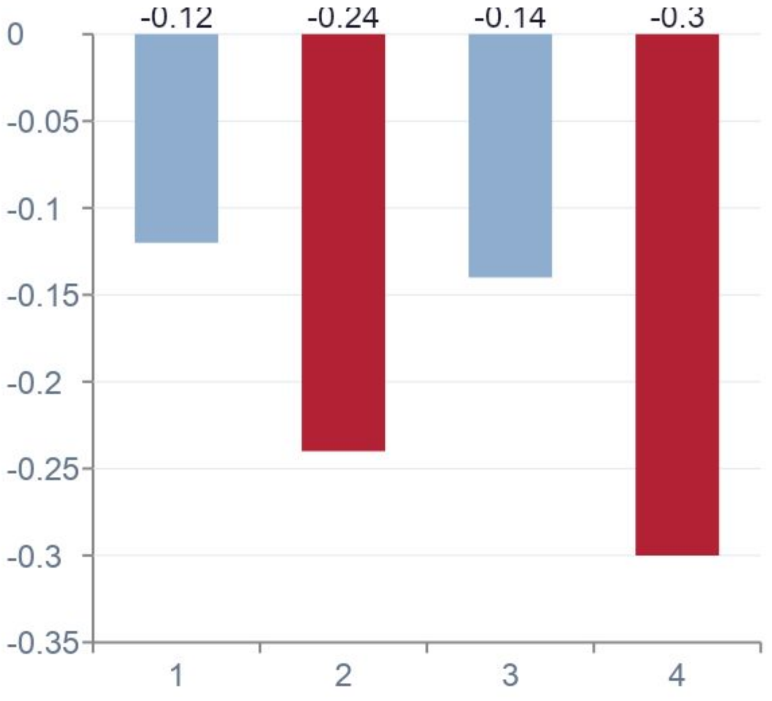
EG = (Wasted Dem − Wasted Rep) / Total Votes

Negative EG → More Democratic votes are wasted → map favors Democrats

Positive EG → More Republican votes are wasted → map favors Republicans

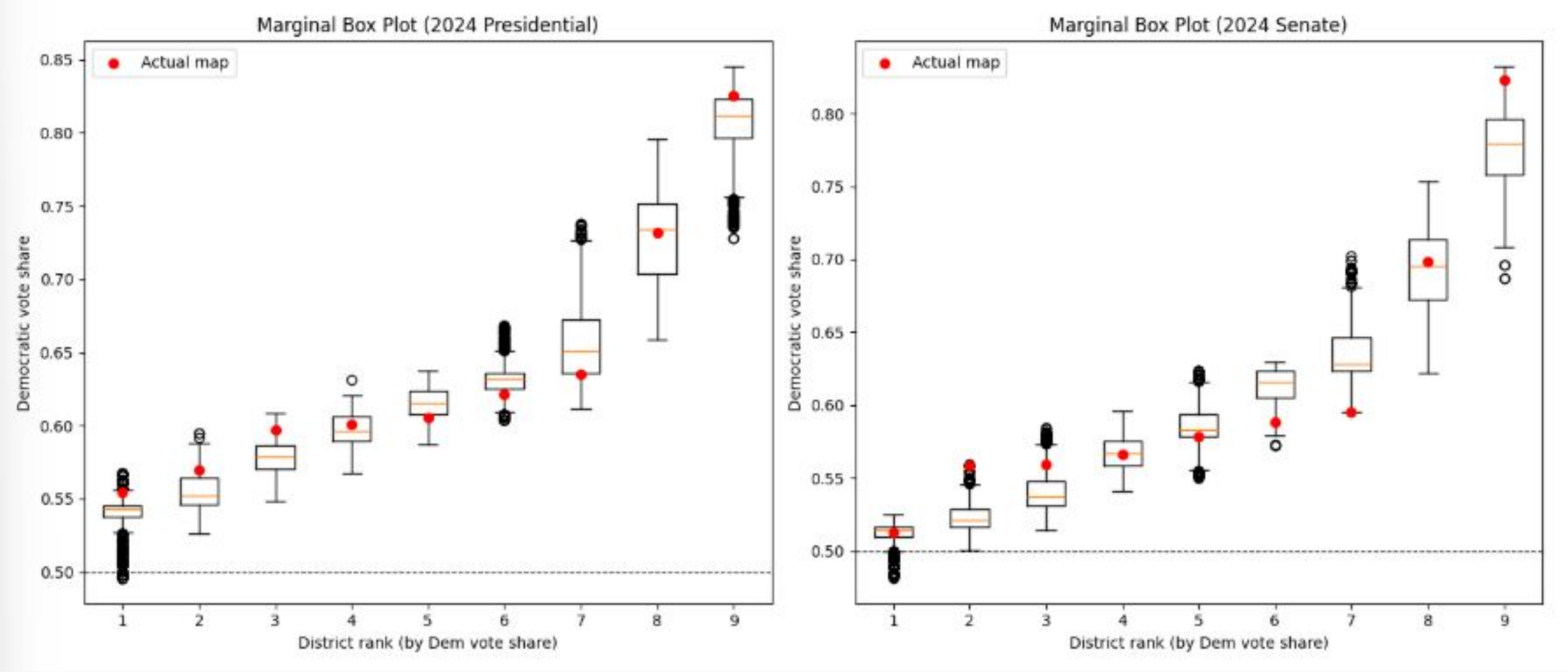
A common threshold for concern is |EG| > 8%. Massachusetts shows EG of −24% (Pres) and −30% (Senate), far exceeding this threshold.

EG Results



Marginal Box Plots: Signature of Gerrymandering

Each box = distribution of Dem vote share across ensemble for each district ranked 1–9. Red dots = actual MA map values.



Key finding: All 9 districts fall above the 50% line in both elections. District 9 is notably packed with Democratic votes compared to the ensemble. Most red dots fall within or near the boxes — the map's district-level margins are not dramatically unusual except at the extremes.

Conclusions

Extreme Efficiency Gap

The actual MA map shows an EG of −24% (Pres) and −30% (Senate), both in the extreme left tail of the ensemble — far more Democratic-favoring than nearly all 2,000 randomly generated maps.

All 9 Districts Democrat

Winning all 9 districts is a rare outcome in the ensemble, suggesting the map is drawn in a way that maximizes Democratic seats.

More Fragmented Borders

659 cut edges vs ensemble mean of ~480 indicates the map has more fragmented district boundaries than typical — consistent with deliberate line-drawing.

District 9 Packing

Marginal box plots show District 9 is packed with Democratic votes beyond what random maps typically produce.